



Anyone who
stops learning
is old, whether
at twenty
or eighty.

—Henry Ford

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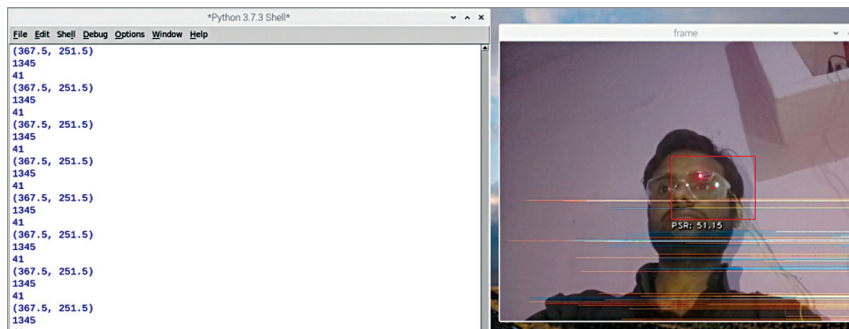


Fig. 5: Author testing the code

```
# Python 2/3 compatibility
from __future__ import print_function
import sys
PY3 = sys.version_info[0] == 3

if PY3:
    xrange = range
    from pynput.mouse import Button, Controller
    from nmap import nmap

import numpy as np
import cv2 as cv
from common import draw_str, RectSelector
import video
from gpiozero import Button as butt
from signal import pause
import mouse
import time
```

Fig. 6: Code snippet

```
prex=0
prey=0
button = butt(27)
mouse = Controller()

mapfnx = nmap(290, 400, 10, 1905, normfn=int)
mapfny = nmap(250, 300, 10, 1059, normfn=int)
```

Fig. 7: Code mapping eye movements to laptop GUI coordinates

```
class MOSSE:
    def __init__(self, frame, rect):
        x1, y1, x2, y2 = rect
        w, h = cv.getOptimalDFTSize, [x2-x1, y2-y1]
        x1, y1 = (x1+x2-w)//2, (y1+y2-h)//2
        self.pos = x, y = x1+0.5*(w-1), y1+0.5*(h-1)
        self.size = w, h
        img = cv.getRectSubPix(frame, (w, h), (x, y))

        self.win = cv.createHanningWindow((w, h), cv.CV_32F)
        g = np.zeros((h, w), np.float32)
        g[h//2, w//2] = 1
        g = cv.GaussianBlur(g, (-1, -1), 2.0)
        g /= g.max()

        self.G = cv.dft(g, flags=cv.DFT_COMPLEX_OUTPUT)
        self.H1 = np.zeros_like(self.G)
        self.H2 = np.zeros_like(self.G)
        for _i in xrange(128):
            a = self.preprocess(rnd_warp(img))
            A = cv.dft(a, flags=cv.DFT_COMPLEX_OUTPUT)
            self.H1 += cv.mulSpectrums(self.G, A, 0, conjB=True)
            self.H2 += cv.mulSpectrums(A, A, 0, conjB=True)
        self.update_kernel()
        self.update(frame)
```

Fig. 8: Some more code

mouse click.

Although this technique is effective, there is one problem. It increases the bill of material to several hundred rupees. Also, people having problem with one eye cannot use this solution.

Therefore, this solution has been modified with the inclusion of a code. Now, if an eye blinks within a second, it is considered as natural. But if an eye blinks for more than a second (say, four seconds), it is considered as intentional.